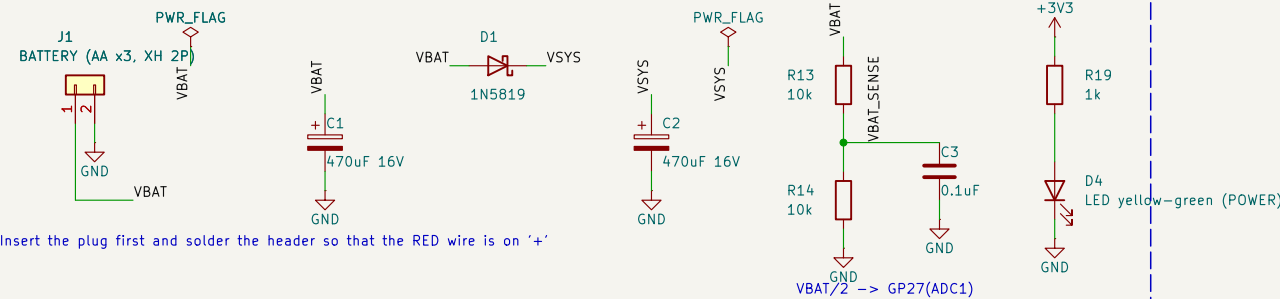


Linetracer2 : Raspberry Pi Pico line tracer for kids (all through-hole)

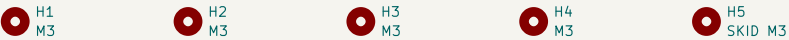
Power: AA x3 (alkaline 4.5V / NiMH 3.6V) -> VBAT (motors) ; VBAT -> D1 Schottky -> VSYS (Pico). USB 5V can not back-feed the batteries. Never use 4 cells (Pico VSYS max 5.5V).

1. POWER

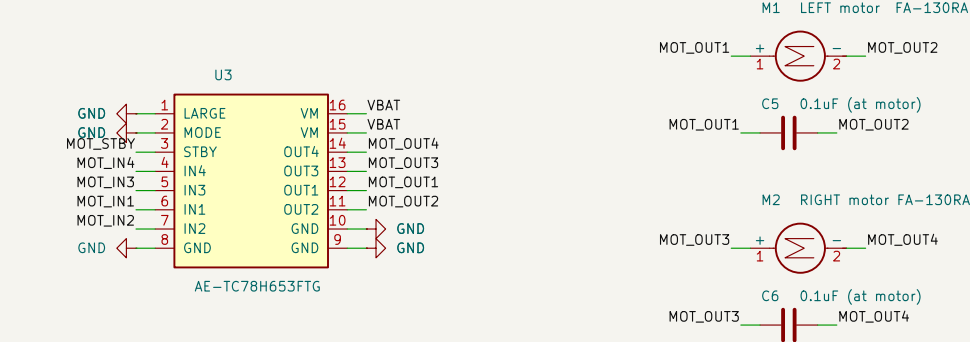


Insert the plug first and solder the header so that the RED wire is on '+'

MECHANICAL: H1-H4 = gearbox frame M3 (x30/x70, y62/y96), H5 = front skid M3

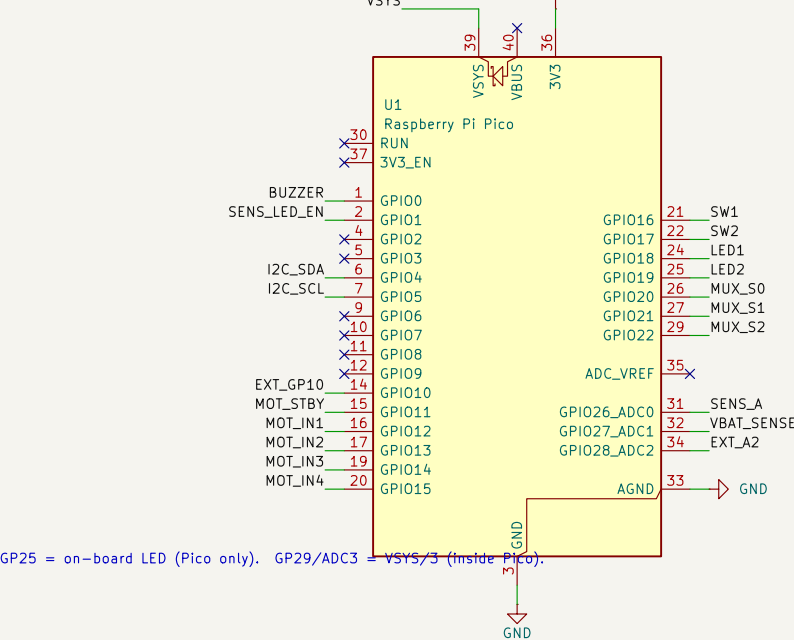


3. MOTOR DRIVER (Akizuki AE-TC78H653FTG module, DIP16)

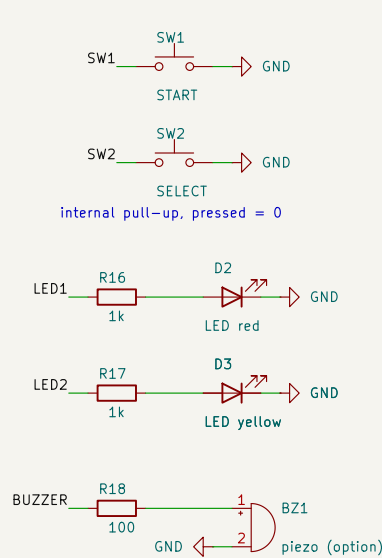


LARGE=L : small mode (2 motors, 2A/ch) MODE=L : IN/IN control
IN1/IN2 -> OUT1/OUT2 = LEFT motor, IN3/IN4 -> OUT3/OUT4 = RIGHT motor
H/L fwd, L/H rev, H/H brake, L/L coast. Over-current (ISD) latches OFF:
toggle STBY L->H to recover (done by the library).

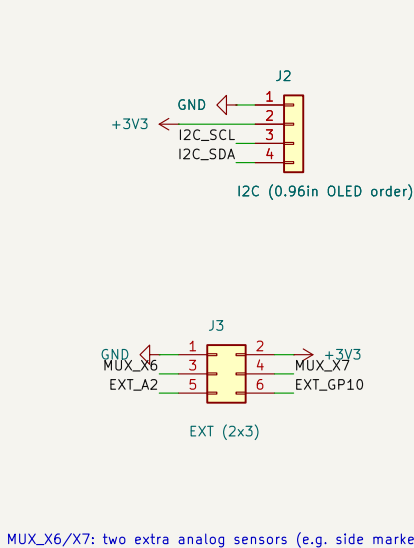
2. MCU Raspberry Pi Pico / Pico 2



4. BUTTONS / LEDs / SOUND

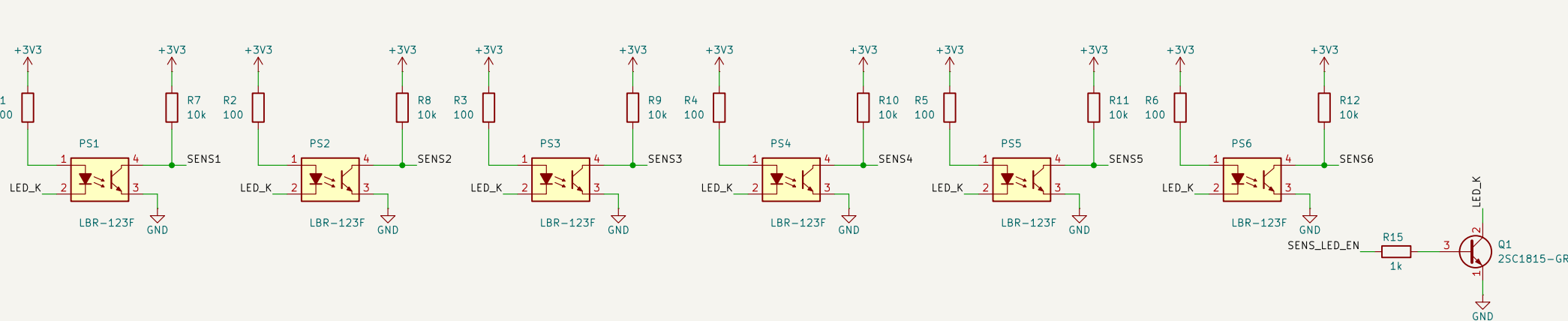


5. EXPANSION (not needed for basic kit)



6. LINE SENSORS 6 x LBR-123F (mount on the BOTTOM side, lens to the floor, 2-3 mm gap)

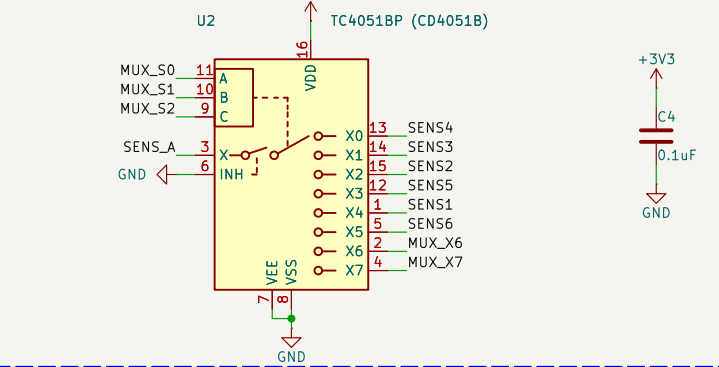
IR LED: (3.3V - 1.25V) / 100 ohm = about 19 mA each. Q1 switches all LEDs (ambient light cancel). Photo transistor: 10k pull-up -> white = low voltage, black = high voltage



PS1 = far LEFT ... PS6 = far RIGHT (seen from above, robot moving up the page)

Q1: all IR LEDs on when SENS_LED_EN = 1 (about 115 mA)

7. ANALOG MUX (6 sensors -> 1 ADC)



See docs/03_circuit.md for the design notes
Motor driver = Akizuki AE-TC78H653FTG module (TC78H653FTG is QFN, pre-mounted)
All parts are through-hole (kids can solder).
EDTC
Sheet: /
File: Linetracer2.kicad_sch
Title: Linetracer2 - kids line tracer (Raspberry Pi Pico + TC78H653FTG)
Size: A3 Date: 2026-10-04 Rev: A
KiCad E.D.A. 10.0.6 Id: 1/1